6CCSACCM: Cooperation, Coordination and Multi-Agent Systems

Week 2

Non-cooperative games (part II)

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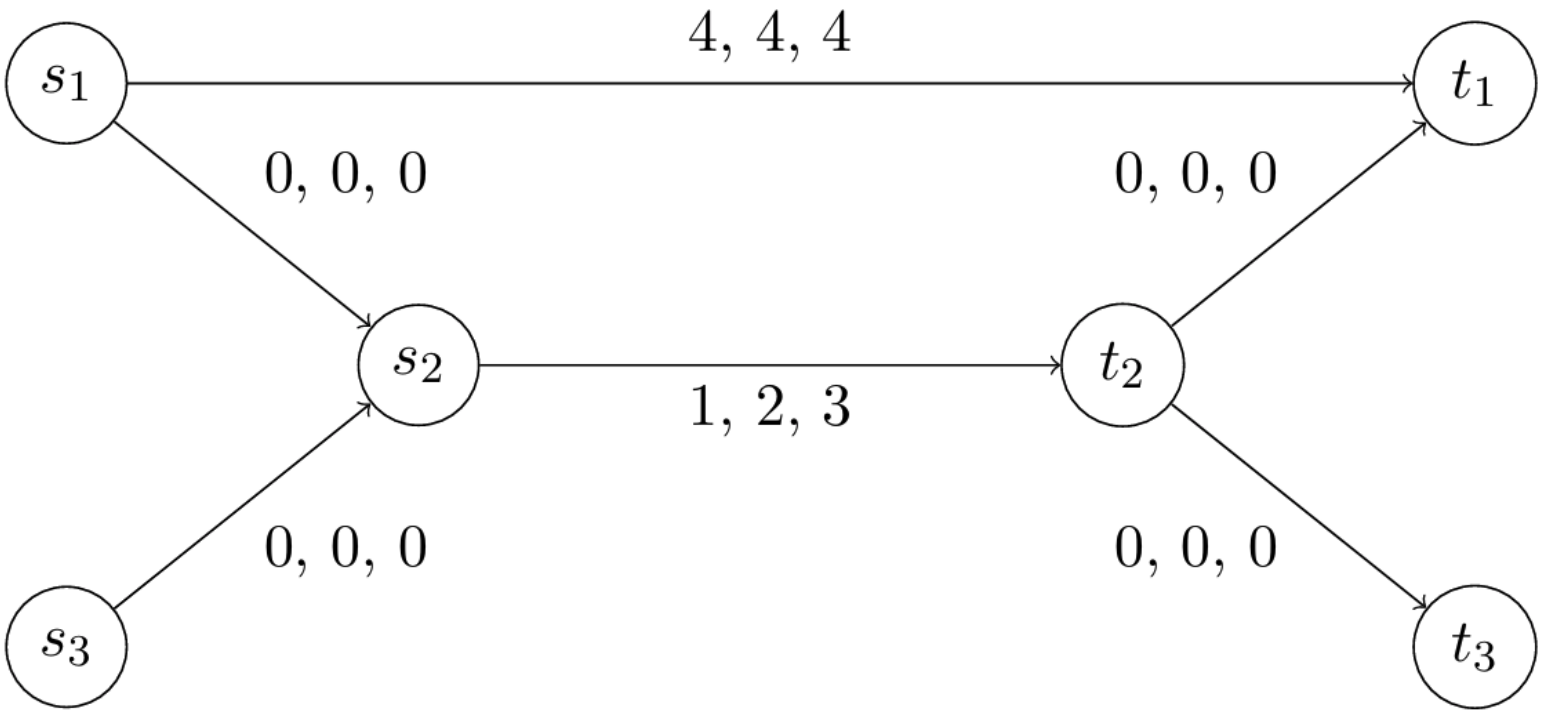
Large games

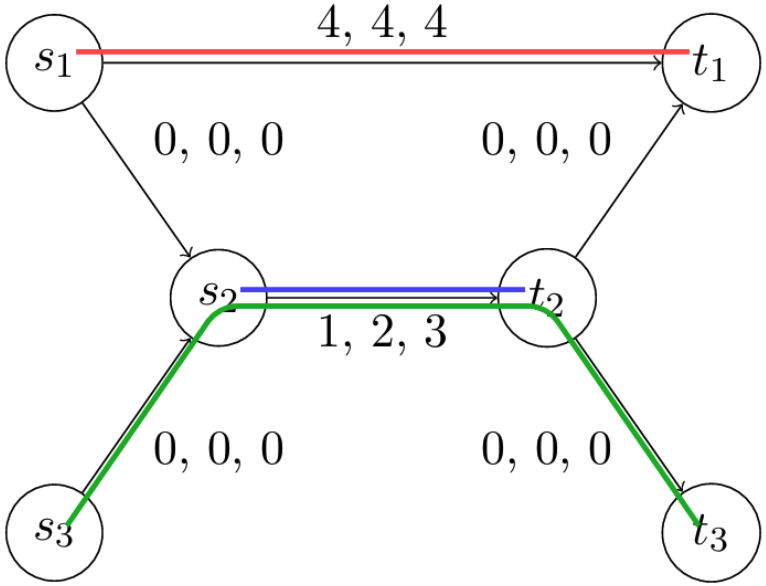
How many numbers do we need
to write down an n-player k-action game?

Example



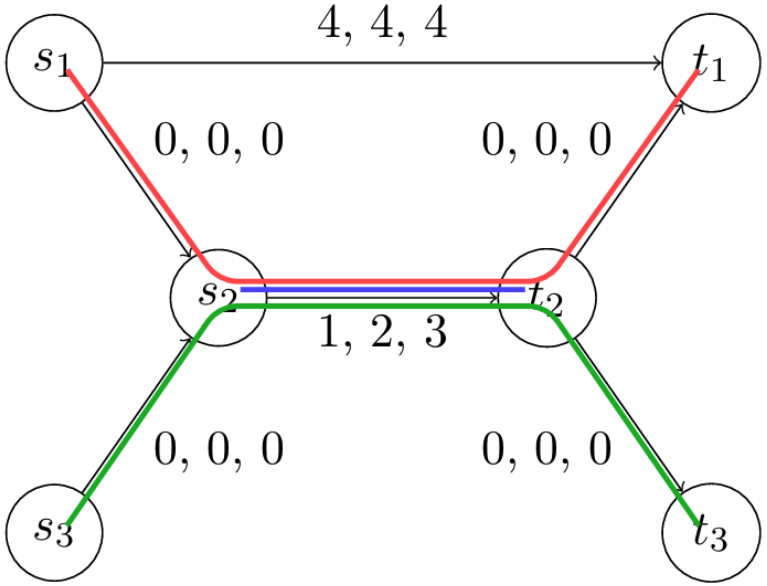
Example

State A:



social cost: $4 + 2 + 2 = 8$

State B:



social cost: $3 + 3 + 3 = 9$

Congestion games

A congestion game is defined by:

- A set of n players N
- A set of m facilities (or, resources) R
- For each player i, a set of actions A_i where each action $a_i \in A_i$ represents a subset of the facilities: $a_i \subseteq R$
- For each facility $j \in R$, a cost (or, delay) function $d_j : \{0, \ldots, n\} \to \mathbb{R}_{\geq 0}$ where $d_j(k)$ is the cost of facility j when k players are using it

Congestion games

Player costs are then defined as follows:

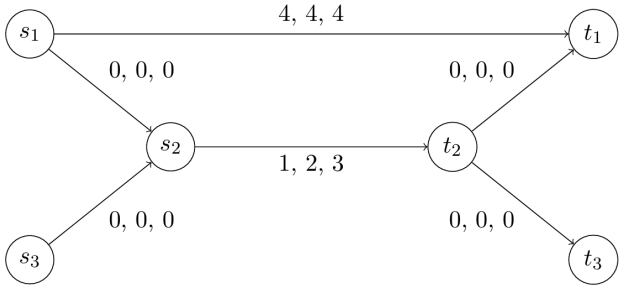
- For action profile $a = (a_1, \ldots, a_n)$ define $n_j(a) = |\{i : j \in a_i\}|$ to be the number of players using facility j.
- Then the cost of agent i is:

$$c_i(a) = \sum_{j \in a_i} d_j(n_j(a))$$

Network congestion (aka routing) games

- In a network congestion game, there is a graph $G = (V, E)$.
- The resource set R corresponds to the set of edges E.
- For each player $i \in N$, there is a dedicated source-target pair (s_i, t_i) such that A_i is the set of paths from s_i to t_i.

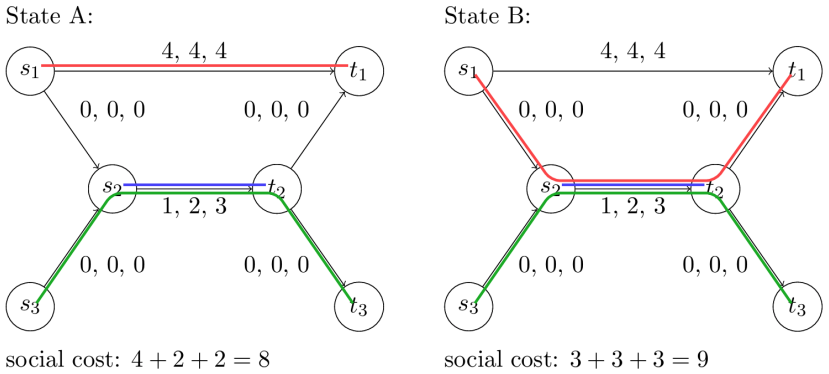
Routing games



In particular, in the above example:

- $N = \{1, 2, 3\}$ and $R = \{(s_1, t_1), (s_1, s_2), (s_2, t_2), (s_3, s_2), (t_2, t_1), (t_2, t_3)\}$
- Player 1's strategy set is given by $A_1 = \{\{(s_1, t_1)\}, \{(s_1, s_2), (s_2, t_2), (t_2, t_1)\}\}$. These are two strategies: The first one uses only a single resource/edge, the second one uses three.
- Players 2 and 3 only have one strategy each.
- The delay function of the resource/edge (s_2, t_2) is $d_{(s_2,t_2)}(x) = x$ for all x.

Routing games



There are two different states:

- We have $n_{(s_2,t_2)}(A) = 2$ and $n_{(s_2,t_2)}(B) = 3$.
- Player 1's cost can be computed as $c_1(A) = d_{(s_1,t_1)}(n_{(s_1,t_1)}(A)) = 4$ in state A and $c_1(B) = d_{(s_1,s_2)}(n_{(s_1,s_2)}(B)) + d_{(s_2,t_2)}(n_{(s_2,t_2)}(B)) + d_{(t_2,t_1)}(n_{(t_2,t_1)}(B)) = 0 + 3 + 0 = 3$.

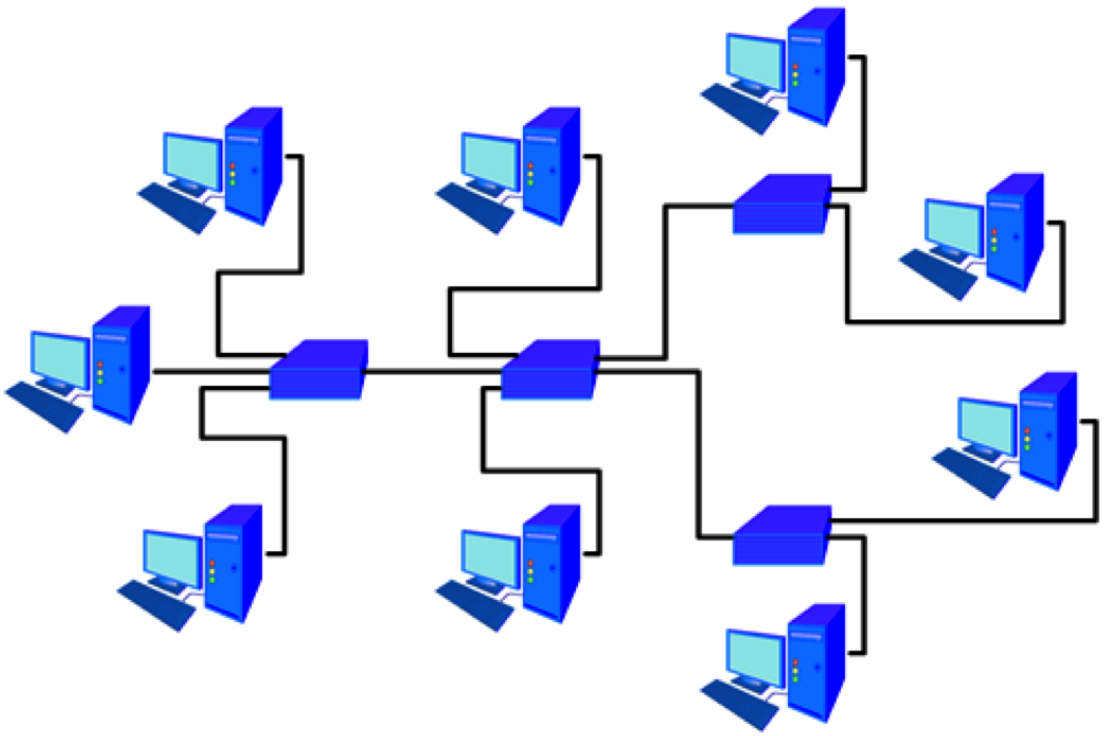
Traffic Routing and Transportation Networks

Congestion games model how drivers or vehicles choose routes, each aiming to minimise travel time. The classic use-case: analysis of road traffic, navigation apps, and highway network design.

Telecommunication and Communication Networks

In wired and wireless networks, data packets (or users) select routes or channels, and performance (latency, bandwidth) decreases with congestion on individual links or frequencies.

- Applications: TCP/IP routing, spectrum allocation, load balancing in cellular networks.



Other applications

Cloud and Distributed Computing

- Tasks or jobs in distributed systems (datacenters, cloud platforms) compete for computational resources (CPUs, memory, bandwidth). Congestion games capture resource sharing and system efficiency.

Multi-Agent Systems and Robotics

- Multiple robots or agents sharing physical infrastructure (warehouse aisles, sensor networks) use congestion games to coordinate movement or task allocation and avoid bottlenecks.

Other applications

Electricity Grids and Smart Energy Systems

- Consumers or smart devices choose when to use electricity, and costs rise with collective demand. Congestion games help analyse demand response and peak shifting.

Internet of Things (IoT) and Sensor Networks

- Devices compete for transmission time, battery usage, or shared sensors. Congestion games inform protocol design for fairness and efficiency.

Other applications

Crowdsourcing and Shared Economy

- Platforms (ride-hailing, shared bikes) allocate resources (drivers, bikes) across a city. Too many users at one location or time reduces utility for all.

Resource Allocation in AI and Machine Learning

- Competing AI agents select computing resources or data sources. Congestion effects arise in federated learning, model ensembling, or online ad auctions.

Other applications

Cybersecurity: Attack and Defense

- Attackers and defenders allocate resources across network surfaces. Over-concentration (congestion) on targets changes payoffs, relevant for intrusion detection and defense games.

Urban Planning and Facility Location

- Citizens or services choose between public resources (parks, hospitals, schools), and overuse degrades quality. Congestion games model optimal placement and usage management.

Best/better response dynamics

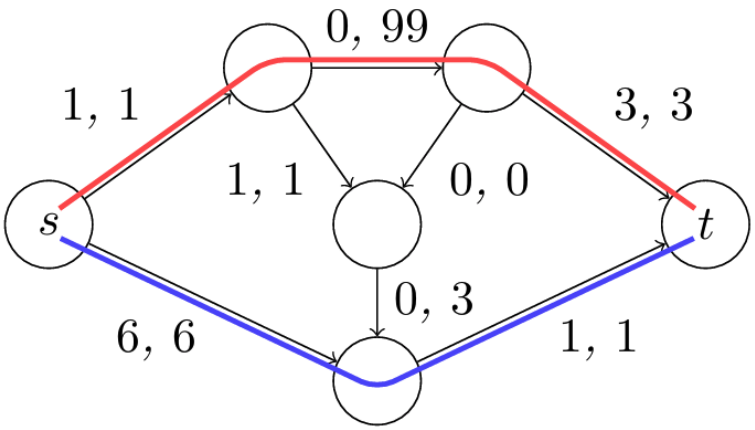
The basic idea:

- Players start playing arbitrary actions
- In arbitrary order, players take turns changing their action if doing so can improve their utility/cost
- Forever...

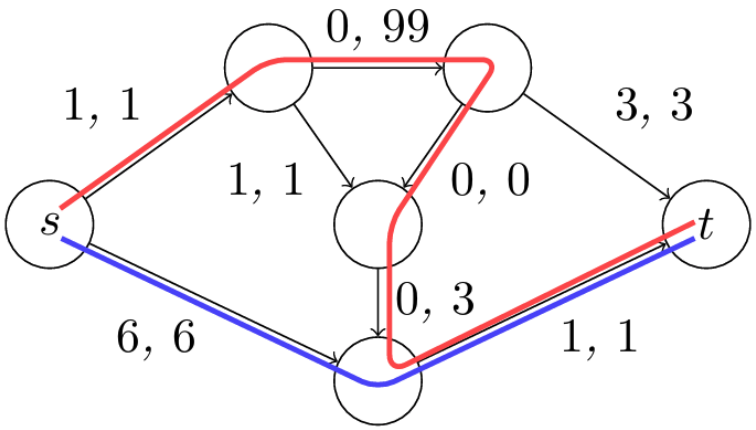
If best response dynamics halts, it returns a pure Nash equilibrium.

Example

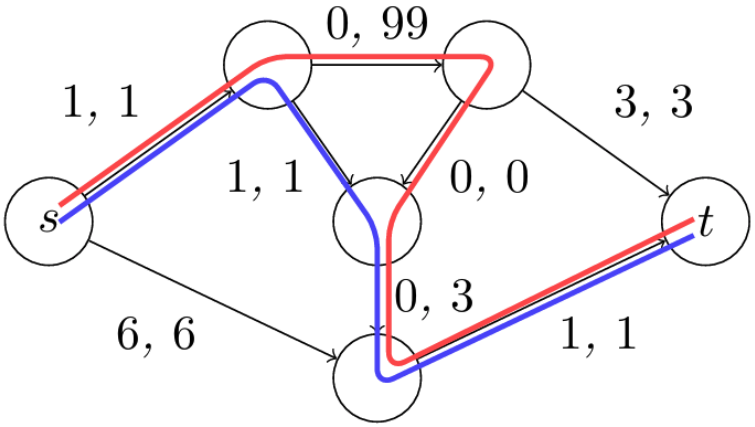
start:



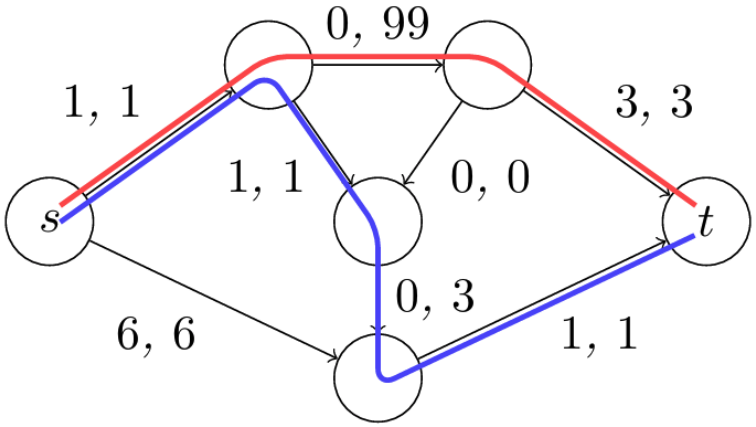
after first improvement (red player):



after second improvement (blue player):



after third improvement (red player):



Best/better response dynamics

Algorithm 1 Best Response Dynamics

Initialize $a = (a_1, \ldots, a_n)$ to be an arbitrary action profile.
while There exists i such that $a_i \notin \arg\min_{a \in A_i} c_i(a, a_{-i})$ **do**
  Set $a_i = \arg\min_{a \in A_i} c_i(a, a_{-i})$
end while
Halt and return a.

Do best response dynamics always halt?

Best/better response dynamics

Theorem (Rosenthal 1973)

For every congestion game, every sequence of improvement steps is finite.

This property is called the **finite improvement property**.

Corollary

Every congestion game has at least one pure Nash equilibrium.

Proof of theorem

The analysis is based on a potential function argument.
For every state a, let

$$\phi(a) = \sum_{j=1}^{m} \sum_{k=1}^{n_j(a)} d_j(k)$$

This function is called Rosenthal's potential function.

Now consider how the potential function changes in a single round of best response dynamics, when player i switches from playing some action $a_i \in A_i$ to playing $b_i \in A_i$ instead.

Proof of theorem

First, because this was a step of best response dynamics, we know that the switch decreased player i's cost:

$$\begin{aligned}
\Delta c_i &= c_i(b_i, a_{-i}) - c_i(a_i, a_{-i}) \\
&= \sum_{j \in b_i \setminus a_i} d_j(n_j(a) + 1) - \sum_{j \in a_i \setminus b_i} d_j(n_j(a)) \\
&< 0
\end{aligned}$$

The change in potential is:

$$\begin{aligned}
\Delta \phi &= \phi(b_i, a_{-i}) - \phi(a_i, a_{-i}) \\
&= \sum_{j \in b_i \setminus a_i} d_j(n_j(a) + 1) - \sum_{j \in a_i \setminus b_i} d_j(n_j(a)) \\
&= \Delta c_i
\end{aligned}$$

Proof of theorem

Hence, we know that $\Delta\phi < 0$. But since ϕ can take on only finitely many different values (why?) and decreases between each round of best response dynamics, best response dynamics must eventually halt (and hence output a pure strategy Nash equilibrium). $\square$

Potential function

A game with player set $N = \{1, 2, \ldots, n\}$, action sets $(A_i)_{i \in N}$, and utility functions $U_i : A \to \mathbb{R}$ is called a **potential game** if there exists a function $\phi : A \to \mathbb{R}$ such that for every action profile $a \in A$, agent $i \in N$, and alternative choice $a'_i \in A_i$,

$$U_i(a'_i, a_{-i}) - U_i(a_i, a_{-i}) = \phi(a'_i, a_{-i}) - \phi(a_i, a_{-i})$$

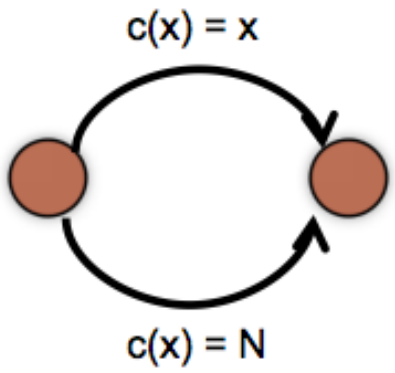
This function ϕ is called the **exact potential function**.

Congestion games are exact potential games.

How bad is selfish routing?

Pigou's example

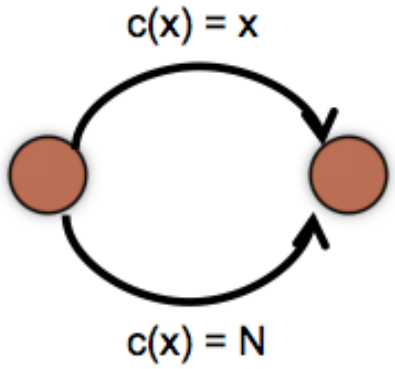
- The cost on the upper link depends on congestion,
- no congestion effects on the lower link.



- What will selfish network users do?
  - Everyone aims to minimise their own cost.

Pigou's example

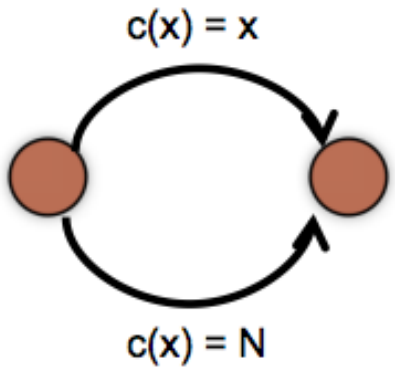
- **Claim**: All traffic will take the top link!



- **Reason**: The bottom traffic is envious!
  - If there are agents on the bottom link, the cost on the top link is $< N$, making the bottom traffic envious.
  - The only stable outcomes are for (almost) all agents to take the top link!

Pigou's example

- **Consider instead**: traffic split equally.



- **Improvement**:
  - half of traffic has cost N, as before,
  - half of traffic has cost $\frac{N}{2}$, twice less than before,
  - the total cost for the system of agents drops from N^2 to $\frac{3}{4}N^2$.

Price of anarchy/stability

- Examines how well an equilibrium solution approximates the optimal solution for a given objective:
  - minimising/maximising the total cost/utility (the *social welfare*);
  - minimising the maximal load/delay (aka *makespan*).

Price of anarchy/stability

- Given an optimisation goal, define the values of an equilibrium and an optimal solutions, with respect to that goal.
- The **Price of anarchy** (PoA) and the **Price of Stability** (PoS) are defined as follows:

$$PoA(S) = \frac{\text{value of worst(best) equilibrium}}{\text{value of optimal solution}}.$$

  - The definition is for the case of minimisation.
  - For a maximisation problem, turn the fraction over, so that $PoA, PoS \geq 1$, anyway.

Prisoner's dilemma

	C	D
C	$10, 10$	$0, 12$
D	$12, 0$	$1, 1$

- There is a single equilibrium, (D, D), while (C, C) maximises the social welfare.
- $PoA = PoS = \frac{20}{2} = 10$.

Battle of sexes

	Theatre	Football
Theatre	$2, 1$	$0, 0$
Football	$0, 0$	$1, 3$

- (Theatre, Theatre) and (Football, Football) are both NE, with values 3 and 4, respectively. The optimal value is 4.
- Thus, $PoS = 1$ and $PoA = \frac{4}{3}$.

Linear congestion games

Theorem (Christodoulou and Koutsoupias, 2005)

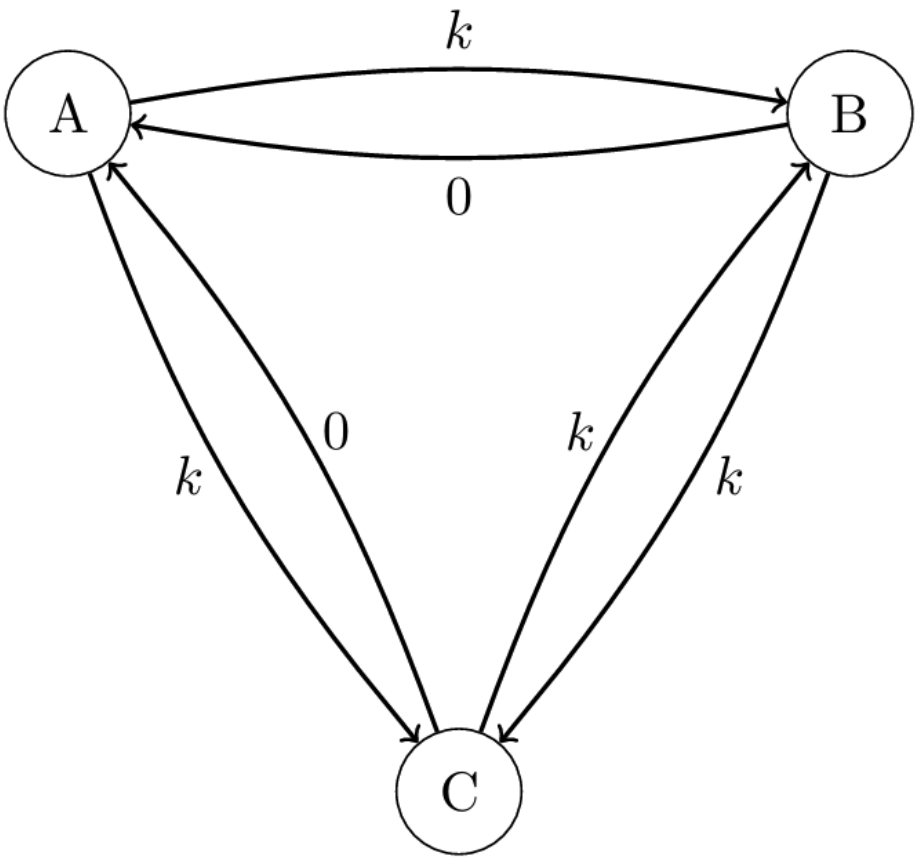
For any congestion game with linear cost functions, the price of anarchy is at most 2.5.

Theorem (Christodoulou and Koutsoupias, 2005)

There are linear congestion games with 3 or more players with price of anarchy for the social cost equal to 2.5.

Example

$$\{(A,B),(A,C),(B,C),(C,B)\}$$



THANK YOU!